

# Nicolas Rohart

Forestry - Design - Robotic Fabrication - Construction

## From standing tree to finished structure.

I operate across the entire timber value chain - forest management and harvesting, computational design, robotic shaping of curved timber, and on-site construction - including the digital workflows that connect each stage.

### BASE

Limousin, France

### EXPERIENCE

15 years, full chain

### LOOKING FOR

R&D partnership

### OPEN TO

Worldwide

### THE CHAIN - FIVE STAGES, ONE HAND THROUGHOUT

#### 01 Forest & Silviculture

Independent forest management, felling and sale, aimed at resilient long-term ecosystems rather than short-term extraction. Developer of "Bois Suivis" - a field-verifiable tracked-timber protocol that goes beyond declared certification schemes.

#### 02 Harvest & Extraction

Felling and skidding done hands-on, in direct feedback with the rest of the chain. Built a custom skidding vehicle from a repurposed 4x4 fire-service chassis - bodywork, crane, structural welding - individually type-approved for road use.

#### 03 Design

Design proposals for custom timber structures from real, round, naturally curved trunks, fitted to each client's brief and site. Rhino 3D for modeling, TwinMotion for renders composited onto photogrammetry scans of the existing sites - near-photorealistic proposals that clients can already picture themselves in.

#### 04 Robotic Shaping

Parametric Grasshopper / KukaPRC workflows generating toolpaths for a 7-axis KUKA robot that follows the tree's actual curvature rather than forcing it into a rectilinear beam. Dedicated 3D scanner: a 3 m trunk becomes a 600,000-point cloud in 3.5 minutes. Carried out at La Manufacture du Châtaignier with mentor Ludovic Mallegol.

#### 05 On-Site Building

Two arrangements available: custom-cut timber pieces to supplied plans, or hands-on involvement through to completion. On-site assembly is carried out with qualified carpenters, coordinating the other trades involved.

### NETWORK

- IBOIS (EPFL Lausanne)
- GC2D (Égletons)
- IUT du Limousin (Limoges)

### WHAT I'M LOOKING FOR

A research team, studio, or workshop working on wood, robotics, or both - bringing 15 years of hands-on chain experience into a lab or production environment as developer or trainer. Based in France, open to relocating.

Robotic fabrication

Timber engineering

Grasshopper / KukaPRC

Forestry & silviculture

Technical training

### PUBLICATION

#### "The tree dictates the shape: reversing the industrial paradigm in rural areas"

L. Mallegol, N. Rohart, N. Champroy - Construction Robotics (Springer), 2026. Scan-to-trajectory robotic shaping of natural curved timber, developed at Coop Monts et Barrages. Presented at RobArch2026, Aarhus, Denmark.

### TECHNICAL PROFILE

#### FORESTRY

Forest assessment

Silviculture

Timber harvesting

#### DIGITAL FABRICATION

Parametric design → toolpath

(Rhino / Grasshopper / KukaPRC)

Commercial renders

(MeshRoom / TwinMotion)

#### CONSTRUCTION

Design

Robot operation

Site installation

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